

DL3004 Final Project

Phase 2 - Physical Data Warehouse Implementation & ETL Pipeline

Objective

Students implement the designed warehouse.

Aligned Labs

- Lab 6: Physical DW Implementation
- Lab 7: ETL & ELT Workflows

1. Data Source Creation

Students can use:

Option A - Public datasets:

- Kaggle
- UCI Repository
- Government open datasets

Option B - Generate synthetic data using Python.

Minimum Size:

- Dimension records: 100+
- Fact records: 10,000+

2. Create Warehouse Architecture

Students implement:

Source Files

↓

Staging Schema

↓

Warehouse Schema

↓

Reporting Schema

PostgreSQL schemas:

- project_staging
- project_dw
- project_reporting

3. Implement Staging Layer

Students create staging tables.

Examples:

- stg_customer
- stg_product
- stg_sales

Apply transformations:

- Remove duplicates
- Handle NULL values
- Standardize text
- Convert data types

Example:

Before:

" lahore "

"LAHORE"

After:

Lahore

4. Implement Dimensions

Example:

dim_customer

Required:

- Surrogate key

Example:

Before:

customer_id = C101

After:

customer_key = 1001

customer_id = C101

5. Implement Fact Loading

Load facts using:

Business key → Surrogate key mapping

Example:

Order CSV

customer_id = C101

↓

dim_customer

↓
customer_key = 1001
↓
fact_sales
↓
customer_key = 1001

6. Data Quality Validation

Students implement checks:

Completeness:

- Missing values

Accuracy:

- Revenue reconciliation

Integrity:

- Foreign key validation

Example:

Fact rows without matching dimension = 0

Phase 2 Deliverables

- PostgreSQL Database SQL scripts
- Staging Schema SQL
- DW Schema SQL
- ETL Scripts SQL/Python
- Data Cleaning Report PDF
- Dimension Load Scripts SQL
- Fact Load Scripts SQL
- Data Quality Report PDF
- GitHub Repository Link

Phase 2 Marks: 40%

Database Implementation: 10

ETL Pipeline: 15

Surrogate Key Implementation: 5

Data Quality Validation: 10

Total: 40